

BPM2_ESOFT QUIZ 2_S2026_GD

of Questions: 30

Question #: 1

A 56-year-old man comes to the physician because of a 3-month history of nausea, weight loss and abnormal bowel movements. Physical examination shows an emaciated patient, with abdominal distension and a palpable mass in the right upper quadrant. CT scan of the abdomen shows a tumor in the right hepatic flexure of the colon. Based on the location of the tumor, which of the following groups of lymph nodes will most likely receive metastasizing cells from the cancerous lesion?

- A. Internal iliac
 - B. Inferior mesenteric
 - C. Celiac
 - D. Superior mesenteric
 - E. Superficial inguinal
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Question #: 2

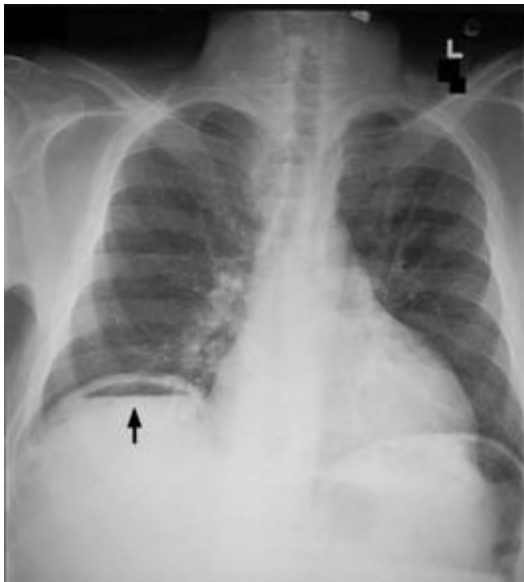
A 30-year-old woman comes to the physician for a routine examination. She is in her third trimester. Ultrasound of the fetus shows herniation of part of the gut tube into the left hemithorax. A diagnosis of congenital diaphragmatic hernia is made. Which of the following is most likely caused by this condition?

- A. Respiratory distress syndrome
 - B. Hemothorax
 - C. Pneumothorax
 - D. Pulmonary hypoplasia
 - E. Eventration of the diaphragm
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Question #: 3

A 25-year-old man is brought to the emergency department because of a penetrating stab wound to the abdomen that he received 30 minutes ago in a brawl. Physical examination shows a large shard of wood penetrating the abdominal wall. A radiograph of the abdomen is attached. Which of the following organs is most likely punctured resulting in the abnormality indicated by the arrow in the image?

- A. Liver
- B. Stomach
- C. Spleen
- D. Pancreas



Question #: 4

A 2-day-old boy is brought to the emergency department by his mother because of severe vomiting after feeding for past 12 hours. Endoscopic studies show an external constriction at the second part of the duodenum. Which of the following is the most likely diagnosis?

- A. Duodenal atresia
- B. Annular pancreas
- C. Duodenal stenosis
- D. Pyloric stenosis

Question #: 5

A 58-year-old man comes to the physician because of a 2-week history of difficulty swallowing. Radiographic studies show an enlarged left atrium impinging on structures in the thorax, especially the esophagus and esophageal plexus. Which of the following structures most likely contains the preganglionic sympathetic fibers that provide innervation to the affected organ?

- A. Vagus nerve
- B. Lateral horn like structures in S2-S4
- C. Lateral horn T1-T4
- D. Paravertebral ganglia T1-T4
- E. Pre aortic ganglion

Question #: 6

An 80-year-old man comes to the emergency department because of a 2-hour history of severe abdominal pain. A CT scan of the abdomen shows ischemic bowel. During laparotomy, a region of ischemic bowel is seen extending from the left colic flexure of the transverse colon to the beginning of the sigmoid colon. Which of the following nervous structures most likely carries the visceral afferent fibers responsible for the patient's pain to the CNS?

- A. Pelvic splanchnics
- B. Lumbar splanchnics
- C. Vagus
- D. Lesser splanchnics
- E. Greater splanchnics

Question #: 7

A 21-year-old woman comes to the emergency department because of a 2-hour history of headache and dizziness. She is at the 17th hour mark of an 18:6 intermittent fasting schedule. She has not had fever, chills, or shortness of breath. She has no personal or family history of major medical illnesses. She has never

smoked cigarettes. Physical examination shows no abnormalities. Laboratory studies show plasma glucose level is 72 mg/dL (70-100 mg/dL). Which of the following is most likely observed in her liver during her current state?

- A. Inhibition of AMP-kinase by glucagon
 - B. Allosteric inhibition of fructose 1,6-bisphosphatase (FBPase-1) activity
 - C. Allosteric activation of pyruvate carboxylase by acetyl-CoA
 - D. Covalent modification and inhibition of glycogen phosphorylase
 - E. High NADH concentration, which activates pyruvate dehydrogenase
 - F. Elevated cytosolic citrate concentration and activation of acetyl-CoA carboxylase
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Question #: 8

An 18-month-old boy is brought to a physician by his parents because of a 6-month history of failure to thrive and tiring quickly when playing. He has had two previous episodes of seizures. He is at the 5th percentile for height and weight. Physical examination shows a protuberant abdomen with a palpable liver margin 2 cm below the right costal margin. Laboratory studies show hyperuricemia, lactic acidemia, elevated serum triglycerides (TAG), and hypoglycemia. Which of the following enzyme deficiency disorders is most likely in this patient?

- A. Phosphofructokinase-1
 - B. Glycogen branching enzyme
 - C. Glycogen debranching enzyme
 - D. Glucose 6-phosphatase
 - E. Glucokinase
 - F. Alpha 1,6-glucosidase
-

Question #: 9

A 10-month-old boy is brought to the emergency department by his parents because of a sudden-onset of seizures. During the past 2 days, he has had fever and cough. Results of laboratory studies obtained on admission are shown in the table. Which of the following disorders is most likely in this patient?

- A. Refsum disease

- B. Zellweger syndrome
- C. Medium chain acyl CoA dehydrogenase (MCAD) deficiency
- D. Myopathic carnitine deficiency
- E. Carnitine palmitoyl transferase - II (CPT-II) deficiency
- F. Glucagon secreting tumor

	Patient values	Reference ranges
Blood glucose	45	70-100 mg/dL
3-hydroxybutyrate	0.3	0.4-0.5 mmol/L

Question #: 10

A 14-year-old boy is brought to the physician because of a 3-week history of muscle cramps. Examination of a muscle biopsy specimen shows normal glycogen levels but increased triglyceride (TAG) droplets. His plasma glucose and ketone body levels are within normal limits. Which of the following processes is most likely defective in this patient?

- A. Oxidation of very long chain fatty acids
- B. Oxidation of branched chain fatty acids
- C. Muscle glycogenolysis
- D. Transport of fatty acids into mitochondria
- E. Endothelial lipoprotein lipase activity
- F. Adipose tissue lipolysis

Question #: 11

A 4-week-old boy is brought to the physician by his mother because of a 5-day history of vomiting and failure to thrive. He is exclusively breast-fed. The mother says that vomiting usually occurred within an hour of breast feeding. Physical examination shows jaundice and an enlarged liver with a palpable lower border of the liver. Examination shows a delay in the developmental milestones. The plasma glucose level is 50 mg/dL. Urinalysis shows the presence of a reducing sugar, which is not glucose. Laboratory studies show an inherited enzyme deficiency. Which of the following reactions is most likely impaired in this patient?

- A. Hydrolysis of lactose to glucose and galactose
 - B. Conversion of fructose to fructose 6-phosphate
 - C. Phosphorylation of galactose
 - D. Transfer of uridyl group to galactose 1-phosphate to form UDP-galactose
 - E. Conversion of fructose 1-phosphate to glyceraldehyde and dihydroxyacetone phosphate
 - F. Oxidation of branched chain fatty acids
-

Question #: 12

A 3-month-old girl is brought to the physician by her mother because of failure to thrive and feeding difficulties since birth. Physical examination shows macroglossia, hepatomegaly and congestive heart failure due to hypertrophic cardiomyopathy. Laboratory studies show elevated serum troponin, CK-MB, ALT and AST, and normal plasma glucose levels. Examination of a muscle biopsy specimen shows the presence of vacuoles that stain positive for a glucose polymer. Which of the following enzyme deficiency disorders is most likely in this patient?

- A. Lysosomal alpha-1, 4-glucosidase
 - B. Glycogen phosphorylase
 - C. Glucose 6-phosphatase
 - D. Glycogen debranching enzyme
 - E. Phosphofructokinase-1
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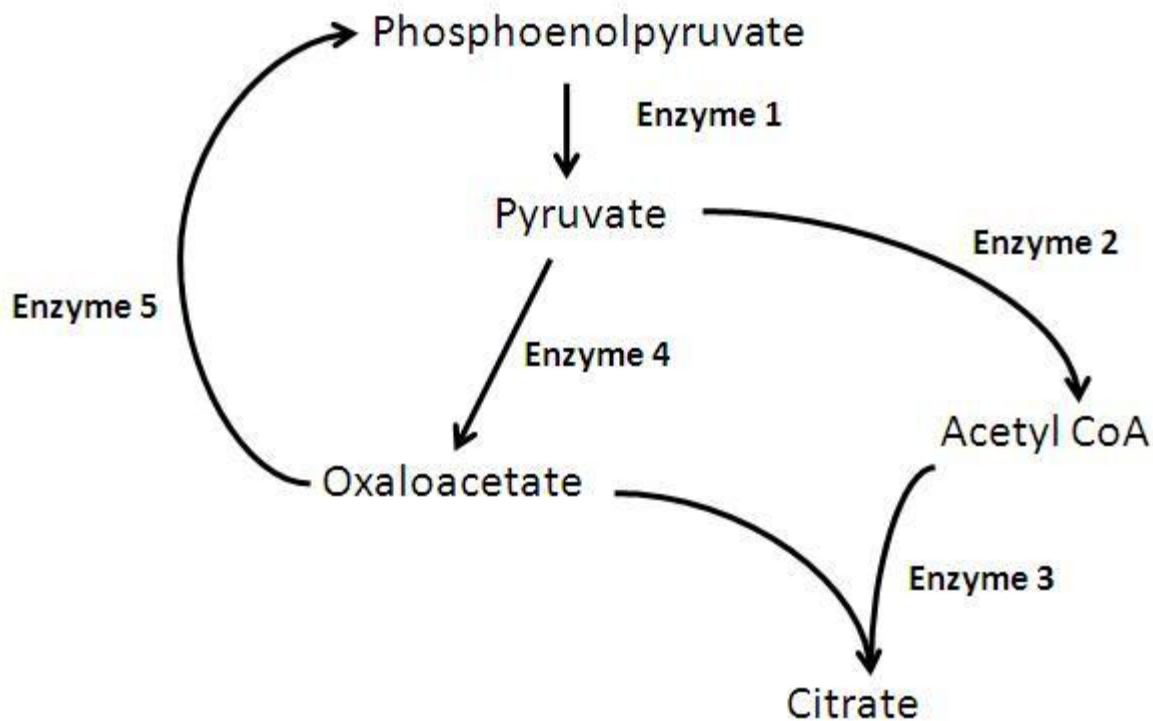
Question #: 13

An investigator studying liver metabolism in experimental rats observes that many hepatic enzymes that are inactivated by phosphorylation when the insulin: glucagon ratio is low. Which of the following enzymes are most likely inactivated by this mechanism?

- A. Enzymes 1 and 2
- B. Enzymes 2 and 3
- C. Enzymes 1 and 4
- D. Enzymes 2 and 5

E. Enzymes 2 and 4

F. Enzymes 1 and 5



Question #: 14

Thirty men aged 30-35 years are recruited for the study of metabolism in the fasting state. It is observed that they have normal blood glucose levels after 24 hours of fasting. Which of the following pathways is most likely responsible for the maintenance of plasma glucose levels at this time?

- A. Glycogen → Glucose 6-phosphate
- B. Oxaloacetate → Citrate
- C. Oxaloacetate → Fructose 1, 6-bisphosphate
- D. Pyruvate → Acetyl CoA
- E. Glucose 6-phosphate → Ribulose 5-phosphate

Question #: 15

A 15-year-old boy is brought to the physician by his parents because of a 1-week

history of unintentional weight loss and polyuria. Laboratory studies show elevated plasma glucose levels and higher-than-normal hemoglobin A1c levels. Urinalysis shows the presence of glucose. Which of the following metabolites is most likely expected to be elevated in this patient's blood and urine?

- A. HMG-CoA
 - B. 3-hydroxybutyrate
 - C. Methylmalonate
 - D. Acylcarnitine derivatives
 - E. α -ketoglutarate
-

Question #: 16

A liver biopsy is performed on volunteers aged 25-30 years for a study on liver metabolism. Following the liver biopsy, a sample of the tissue is homogenized and then centrifuged to remove membrane and organelle material, leaving only cytosolic material in the supernatant. Which of the following enzymes is most likely present in the supernatant?

- A. Acetyl CoA carboxylase
 - B. Medium chain acyl CoA dehydrogenase
 - C. 3-Hydroxy-3-methylglutaryl-CoA lyase
 - D. Ornithine transcarbamylase
 - E. Carnitine palmitoyl transferase-II
 - F. Pyruvate dehydrogenase complex
 - G. Fo-F1 ATP synthase
 - H. Succinate dehydrogenase
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Question #: 17

A 4-month-old girl is brought to the emergency department by her parents because of a sudden onset of seizures. Results of laboratory studies obtained on admission are shown. Which of the following inborn errors of metabolism is most likely in this patient?

- A. von Gierke disease
- B. Pyruvate dehydrogenase deficiency
- C. Classical galactosemia
- D. Glucokinase deficiency
- E. Hereditary fructose intolerance
- F. Glucose 6-phosphate dehydrogenase deficiency
- G. Pyruvate kinase deficiency

	Patient's values	Reference range
Arterial pH	7.2	7.36-7.44
Blood glucose	88	70-90 mg/dL
Serum lactate	6	0.4-1.7 mmol/L
Serum pyruvate	0.5	0.08-0.16 mmol/L

Question #: 18

A researcher studying metabolism observes that fatty acid de novo synthesis and cholesterol synthesis are active in the liver after a carbohydrate-rich meal. Which of the following enzymes most likely provides the reducing equivalents (NADPH) for both pathways?

- A. HMG-CoA reductase
- B. Acetyl-CoA carboxylase
- C. Glucose 6-phosphate dehydrogenase
- D. Glyceraldehyde 3-phosphate dehydrogenase
- E. HMG-CoA synthase
- F. Pyruvate dehydrogenase complex

Question #: 19

A 12-year-old boy is brought to the emergency department by his father after

becoming less responsive following several bouts of nausea and vomiting. Laboratory studies show elevated serum acetoacetate and 3-hydroxybutyrate and ketone bodies are present in the urine. Which of the following enzymes is most likely required for the synthesis of these compounds in the liver?

- A. Succinyl CoA: Acetoacetate CoA transferase
 - B. HMG CoA reductase
 - C. Hormone sensitive lipase
 - D. Lecithin cholesterol acyl transferase
 - E. Mitochondrial HMG CoA synthase
 - F. Acetyl CoA carboxylase
-

Question #: 20

A 20-year-old man comes to the physician because of a 1-day history of passing brown colored urine, muscle pain and muscle cramps after running on the treadmill for about 30 minutes. He says he can tolerate mild exercise. Examination of a muscle biopsy specimen shows excessive glycogen accumulation. Muscle triglyceride levels are within reference range. Examination of a liver biopsy specimen shows normal glycogen structure and content. Which of the following additional laboratory findings will most likely be seen in this patient?

- A. Low blood lactate levels following exercise
 - B. Fasting hypoglycemia
 - C. Presence of reducing sugar in urine
 - D. Decreased muscle carnitine levels
 - E. Increased serum C-10 acyl carnitine levels
-

Question #: 21

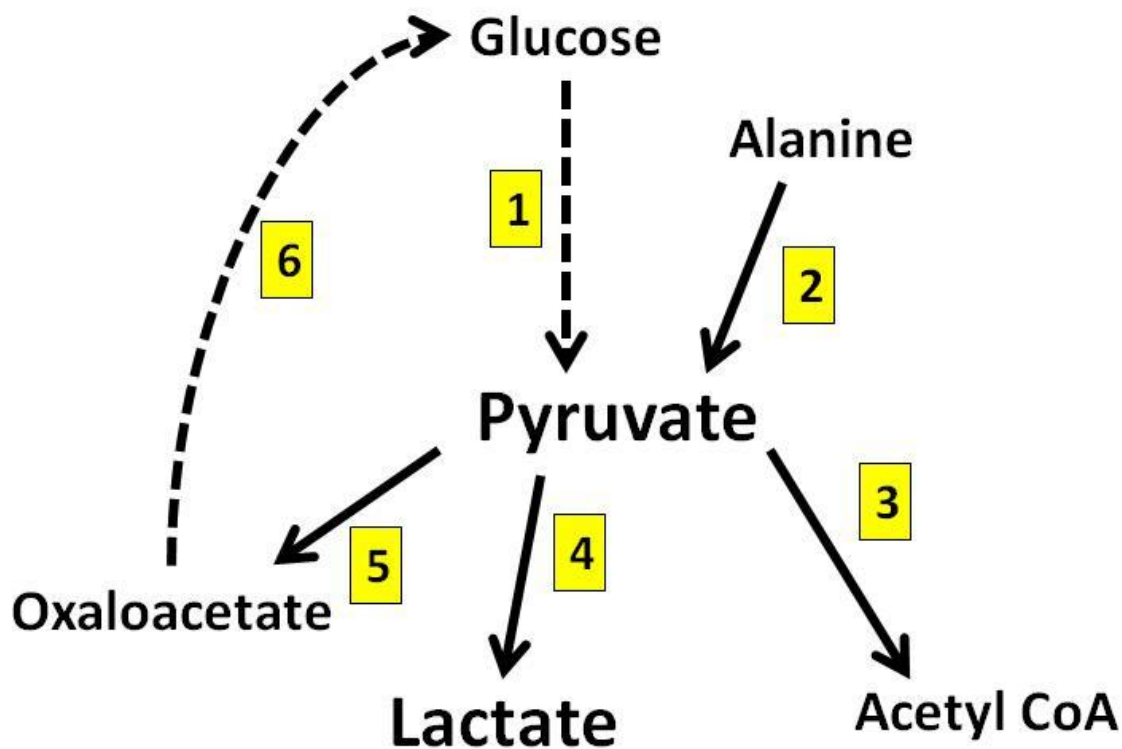
A 13-month-old girl is brought to the physician by her parents for a follow-up examination. She has a 2-month history of episodes of lethargy that occurs a few hours after the consumption of fruits. Blood glucose levels during these episodes ranged from 45-50 mg/dL (reference range: 70-110 mg/dL). Physical examination shows mild icterus and hepatomegaly. She is advised to avoid a specific carbohydrate in her diet. Which of the following reactions is most likely impaired in this patient?

- A. Cleavage of fructose 1-phosphate
 - B. Formation of UDP-galactose
 - C. Phosphorylation of galactose
 - D. Phosphorylation of fructose
 - E. Substrate level phosphorylation in glycolysis
 - F. Intestinal digestion of lactose
-

Question #: 22

An investigator studying the role of pyruvate in intermediary metabolism identifies various reactions by which it is formed and utilized in various tissues and organs. A flow chart representing its metabolism is shown in the figure. Which of the following statements best describes the pathway shown?

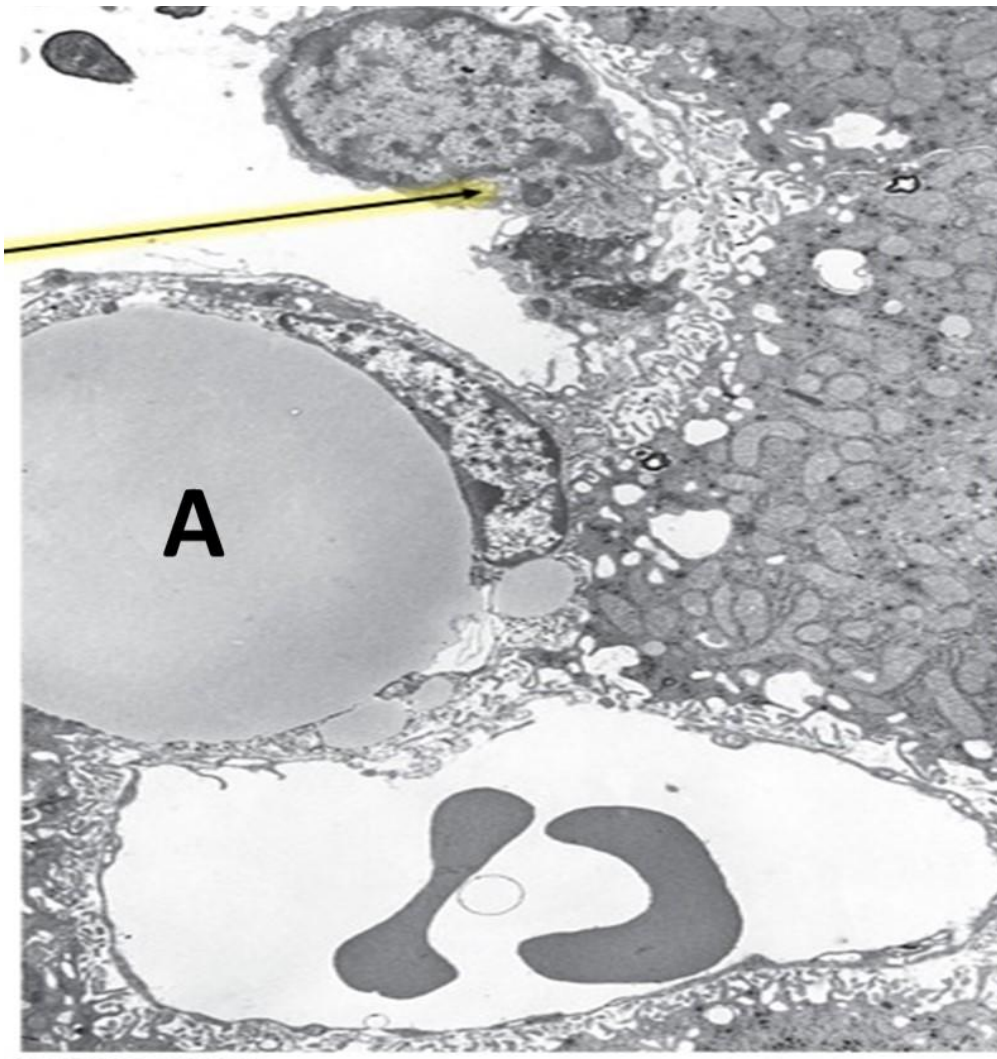
- A. Pathway 1 is inhibited by high concentrations of fructose 2,6-bisphosphate in the liver
- B. Step 5 is inhibited by high levels of acetyl CoA
- C. Step 4 occurs in the actively contracting skeletal muscle
- D. Biotin deficiency results in low activity of Enzyme 4
- E. Pathway 6 is inhibited by glucagon
- F. Enzyme catalyzing step 2 requires TPP as a coenzyme



Question #: 23

A 73-year-old man comes to the physician because of a 6-week history of right sided upper abdominal pain and yellowing of the skin. Physical examination shows an enlarged liver with jaundice. An ultrasound-guided biopsy is obtained from the liver and a TEM of a hepatic sinusoid is shown in the attached micrograph. Which of the following is most likely a function of the cell shown at the label **'A'**?

- A. Vitamin A storage
- B. Sense bile composition
- C. Phagocytosis
- D. Bile production
- E. Secretion of nitric oxide



Question #: 24

A 32-year-old man comes to the physician because of a 6-month history of intermittent abdominal cramps and bloating. History shows the symptoms typically appear following a meal rich in carbohydrates. Physical exam shows no abnormalities. Genetic testing shows a mutation in a gene for an enzyme that breaks down oligosaccharides to monosaccharides for absorption. Which of the following best characterizes the localization of this enzyme?

- A. Digestive organelles of hepatocytes
- B. Cytoplasm of enterocytes
- C. Glands of the intestinal submucosa
- D. Lumen of the stomach

Question #: 25

A 61-year-old man undergoes surgery to remove a peptic ulcer of his stomach. As a result of the surgery, a large number of cells with a basal basophilia usually found at the fundus of the gastric glands are lost. The loss of these cells will most likely cause a decrease in which of the following products?

- A. Pepsinogen
 - B. Secretin
 - C. Lysozyme
 - D. Mucus
 - E. Intrinsic factor
-

Question #: 26

A 51-year-old man is brought to the emergency department because of a 3-hour history of severe abdominal pain. He has been binge drinking alcohol (consumption of excessive amounts of alcohol) over the past 2 days. Physical examination shows acute tenderness to his epigastric region. Laboratory studies of the blood show elevated levels of pancreatic enzymes suggestive of acute pancreatitis. This condition is characterized by autolysis of the secretory units of the parenchyma of the gland. Which of the following hormones stimulate secretion of these enzymes?

- A. Gastrin
 - B. Serotonin
 - C. Cholecystokinin
 - D. Motilin
 - E. Pepsin
-

Question #: 27

A 51-year-old alcoholic man is brought to the emergency department because of a

1-hour history of severe vomiting of bright red blood. Investigation lead to the diagnosis of esophageal varices secondary to portal hypertension. Which of the following is more than likely a feature of the layer of the wall of the esophagus where these varices are usually located?

- A. Stratified squamous epithelium
 - B. Myenteric plexus
 - C. Mucous glands
 - D. Peyer's patches
 - E. Smooth muscle bands
-

Question #: 28

A 55-year-old man is brought to the emergency department because of a 1-month history of worsening abdominal pains, constipation and a weight loss of 6.8-kg (15-lb). Rectal examination shows a mass in the distal rectum that bleeds easily. A biopsy of the mass shows cells with multiple mitoses and loss of architecture. Which of the following is most likely a normal histological feature of the structure in which the mass was found?

- A. Well-developed brush border
 - B. Intestinal villi
 - C. Numerous goblet cells
 - D. Submucosal glands
 - E. Paneth cells
-

Question #: 29

A 45-year-old man comes to the physician because of a 2-month history of headaches, muscle cramps, and fatigue. His blood pressure is 160/110 mm Hg, pulse is 84/min, and temperature is 37°C (98.6°F). Physical examination shows a loud S2 in the right upper sternal border. Laboratory studies show:

Test	Patient value (Reference range)
Serum sodium	145 mEq/L (Normal: 135-145 mEq/L)
Serum potassium	2.6 mEq/L (Normal: 3.5-5.5 mEq/L)

Test	Patient value (Reference range)
Serum bicarbonate	40 mEq/L (Normal: 22-24 mEq/L)
Plasma aldosterone	80 (Normal: 10-30 ng/dL)

An ECG shows flattened T waves. Ultrasonography of the abdomen shows no abnormalities. Which of the following plasma assays is most likely to confirm the diagnosis in this patient?

- A. Catecholamines
- B. Dehydroepiandrosterone
- C. Renin
- D. ACTH
- E. Cortisol

Question #: 30

A 14-year-old girl is brought to the physician by her mother because of a 10-month history of progressive increase in height, increase in shoe size, and weight gain. The mother says that her daughter is growing taller than her friends. Menarche occurred 3 months ago. She is 6 ft (183 cm) tall and weighs 86 kg (190 lb). Her blood pressure is 140/88 mm Hg; other vital signs are within normal limits. Physical examination shows increased thickness of fingers and toes, coarse skin, and prominent mandible and maxilla. Laboratory studies show elevated random blood glucose and plasma IGF-1. Which of the following is the most likely diagnosis?

- A. Acromegaly
- B. Cushing's disease
- C. Addison's disease
- D. Levi loran disease
- E. Gigantism